

# Philip Cho

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## Summary

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Machine Learning Engineer and founder with 5+ years of experience owning the full ML product lifecycle, from research and experimentation to cloud-native deployment, MLOps, and scalable production systems. Specialized in computer vision, applied ML, synthetic data generation, LLM inference, and Kubernetes infrastructure. Strong background translating research into productized platforms, building automated data pipelines, and deploying reliable ML systems on GCP and modern cloud infrastructure.

## Founder Experience

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### Founder, Lead Engineer

*Sota Vault (sotavault.ai)*

Jan 2026 – Present

Vancouver, BC

- Built and launched a full-scale automated machine learning benchmark and leaderboard platform from scratch, owning the product architecture and lifecycle from data ingestion to production cloud deployment.
- Designed cloud-native infrastructure on Google Kubernetes Engine (GKE) with Terraform, including horizontal pod autoscaling for fluctuating user traffic, Redis for API caching, and a GitHub Actions CI/CD pipeline that tests, packages images to Artifact Registry, and finally deploys through Cloud Deploy.
- Developed an automated arXiv ingestion pipeline, first using OCR models to collect metadata and processing key metrics using AWQ/GGUF quantized vLLM models, finally cross checking the quality of the data using web-grounded LLM agents and internal vector DB.
- Architected low-latency semantic search and retrieval with vector search in managed Firestore, enabling accurate model comparison and discovery.

## Work Experience

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### Machine Learning Software Developer II

*Captura (Skylab Technologies before acquisition)*

June 2024 – Jan 2026

Vancouver, BC

- Took ownership of the end-to-end color enhancement initiative, leading research, experimentation, model design, validation, and production rollout for a studio-grade image enhancement pipeline.
- Replaced a legacy multi-output regression approach with a self-supervised color transfer system using reusable presets and contrastive learning, reducing per-studio training data needs to roughly 1% while preserving output quality.
- Drove deployment and operational improvements across the ML pipeline, including model packaging, inference optimization, validation workflows, and CI/CD-friendly delivery practices for repeatable releases.
- Modernized face detection by migrating from a legacy GPU-based model to an accelerated CPU-based service, reducing hallucinations and materially lowering cloud infrastructure cost.

### Machine Learning Software Developer I

*Skylab Technologies*

Sep 2022 – June 2024

Vancouver, BC

- Led ML infrastructure improvements across the full pipeline, refactoring the central codebase and consolidating fragmented training datasets into a unified database and codebase to improve reliability and training throughput.
- Owned the patch-based GAN stray-hair removal project from research through prototyping and dataset strategy, building extraction and synthesis workflows to generate targeted synthetic training data.
- Strengthened MLOps practices around data organization, reproducibility, and release readiness, helping scale model development and contributing to the company's successful acquisition in November 2024.

### Computer Vision Software Engineer

*OneCup AI*

May 2021 – Aug 2022

Vancouver, BC

- Built and deployed an end-to-end ML video processing backend using Nvidia DeepStream and Triton Inference Server

to run real-time inference on livestock surveillance footage.

- Took charge of the synthetic data pipeline by developing Unreal Engine environments to generate training data and reduce data scarcity in production computer vision workflows.
- Applied Conditional GAN techniques to narrow synthetic-to-real domain gap, improving model generalization and supporting more robust deployment performance.
- Collaborated across research and engineering to operationalize computer vision models with production-minded pipeline design, deployment workflows, and system monitoring considerations.

## Customer Success Specialist

*Wishpond*

Oct 2016 – Feb 2019

Vancouver, BC

- Partnered closely with the development team to perform daily QA checks, validate bug reports, and monitor platform behavior, building an early foundation in product quality and technical problem solving.
- Investigated fraud and dispute claims end to end by coordinating with cardholders, financial institutions, and internal stakeholders, collecting case evidence and executing chargeback workflows with accuracy.
- Analyzed customer satisfaction survey trends and dispute patterns, then influenced management to improve refund-related terms and conditions so they were clearer, fairer, and easier for customers to understand.
- Supported VIP customers through consultations, email follow-ups, and webinars, combining hands-on product knowledge with a strong interest in helping users succeed with software.
- Improved onboarding by creating and updating tutorial content across the knowledge base, YouTube channel, video course pages, and chat support resources, including filming in-app training videos for newly released features.

## Education

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### Master of Computer Science in Visual Computing

*Simon Fraser University*

Sep 2019 – Sep 2021

Vancouver, BC

- Completed a machine learning-focused graduate program, studying computer vision, 3D graphics/vision, AR/VR, and natural language processing, achieving above-average academic standing.

### Bachelor of Commerce

*University of British Columbia, Sauder School of Business*

Sep 2011 – May 2016

Vancouver, BC

- Graduated with a dual major in Marketing and International Business.

## Certifications

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### Google Cloud Professional Cloud Architect (PCA)

*Google Cloud*

May 2026

Cloud Architecture and MLOps

- Validated expertise in designing secure, scalable, and reliable GCP architectures, with emphasis on cloud-native ML systems, production infrastructure, and end-to-end MLOps workflows.

## Skills

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**Programming:** Python, JavaScript, SQL

**ML & AI:** PyTorch, Scikit-learn, Keras, OpenCV, Hugging Face, OpenAI, LangChain, computer vision, self-supervised learning, contrastive learning, GANs

**LLM & Retrieval:** vLLM, Qwen, DeepSeek, AWQ/GGUF quantization, LLM agents, OCR pipelines, vector search, semantic retrieval, benchmark evaluation

**MLOps & Cloud:** GCP PCA, GKE, Firestore, GCS, GCE, Pub/Sub, GitHub Actions, Terraform, Artifact Registry, Cloud Deploy, HPA, model deployment, ML pipelines

**AWS & Platforms:** S3, EC2, ECS, SQS, Lambda, Rekognition, Nvidia DeepStream, Triton Inference Server, Unreal Engine

**Specialties:** Benchmark platforms, color enhancement, face detection, video analytics, synthetic data generation, domain adaptation, end-to-end ML systems